7-ordering: 序关系 (Ordering)

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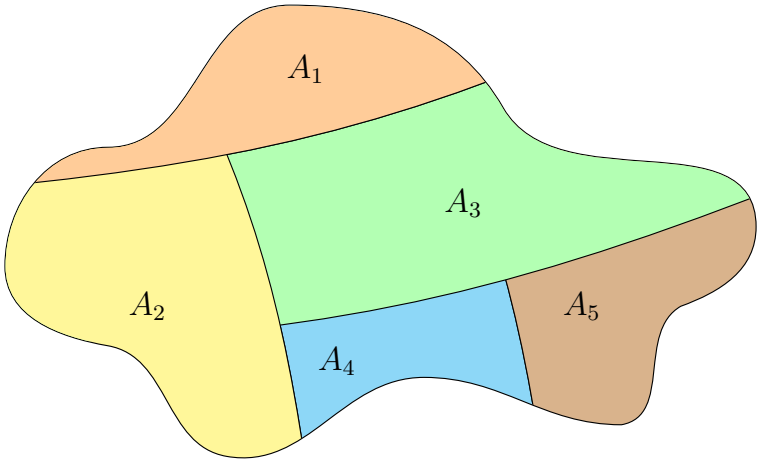
2026 年 04 月 16 日

Definition (Equivalence Relation)

$R \subseteq X \times X$ is an *equivalence relation* on X iff R is

- reflexive (自反): $\forall a \in X.\ aRa$
- symmetric (对称): $\forall a, b \in X.\ (aRb \leftrightarrow bRa)$
- transitive (传递): $\forall a, b, c \in X.\ (aRb \wedge bRc \rightarrow aRc)$



Equivalence Relation $\Longleftrightarrow$ Partition $\Longleftrightarrow$ Abstraction

Definition (Equivalence Relation)

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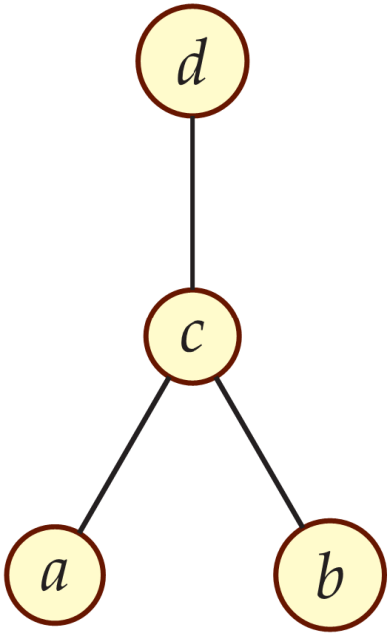
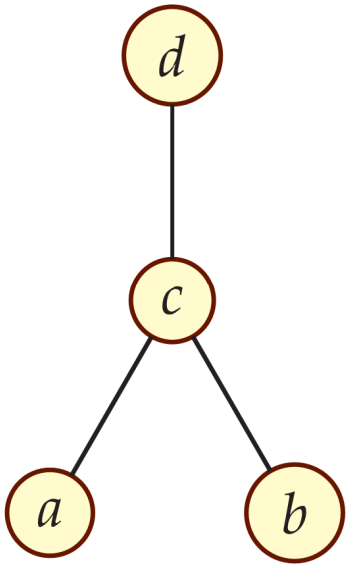


Ordering?

"Order theory is a branch of mathematics that investigates the intuitive notion of order using binary relations.



It provides a formal framework for describing statements such as "this is less than that" or "this precedes that"."

— *Order theory (Wikipedia)*

Definition (偏序关系 (Partial Order; Poset))

令 $\preceq \subseteq X \times X$ 是 X 上的二元关系。

如果 $\preceq$ 满足以下条件, 则称 $\preceq$ 是 X 上的偏序关系,

并称 $(X, \preceq)$ 为偏序集 (poset; Partially Ordered Set):

(1) $\preceq$ 是自反 (reflexive)的。

$$\forall x \in X \; x \preceq x.$$

(2) $\preceq$ 是**反对称 (antisymmetric)**的。

$$\forall x, y \in X \; x \preceq y \wedge y \preceq x \to x = y.$$

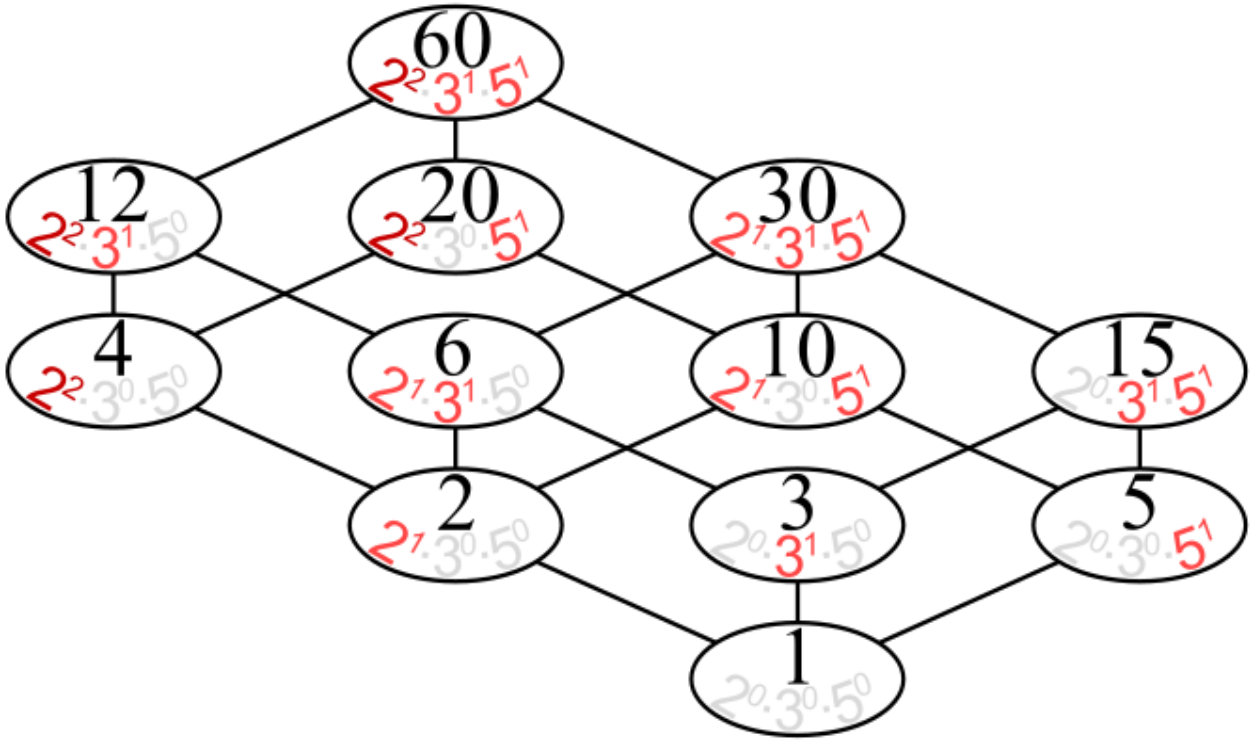
(3) $\preceq$ 是传递 (transitive)的。

$$\forall x, y, z \in X \; x \preceq y \wedge y \preceq z \to x \preceq z.$$

$$x \prec y \triangleq x \preceq y \wedge x \neq y$$

$$X = \{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}$$

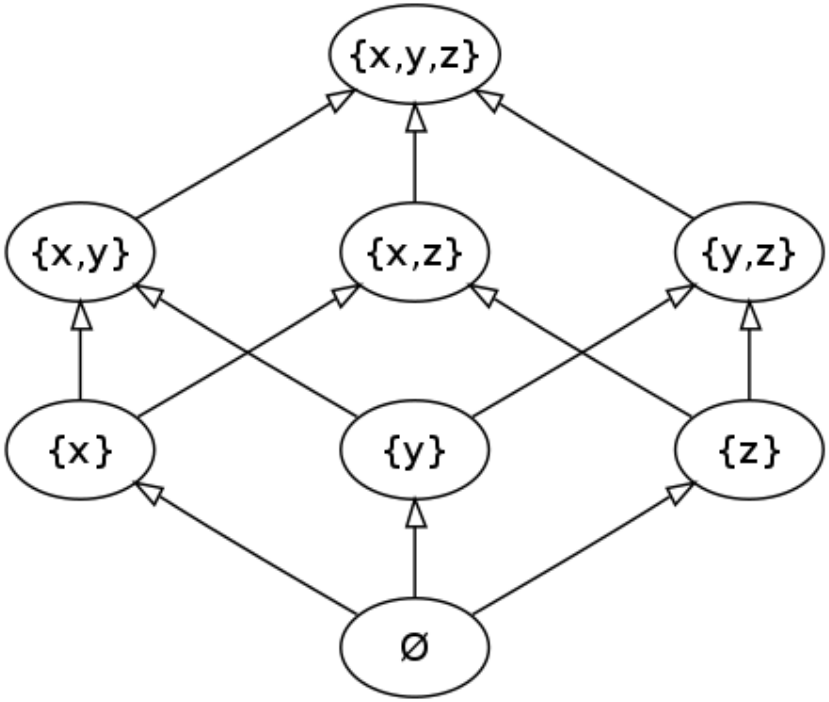
X 上的整除关系



自反 + 反对称 + 传递

$$S = \{x, y, z\}$$

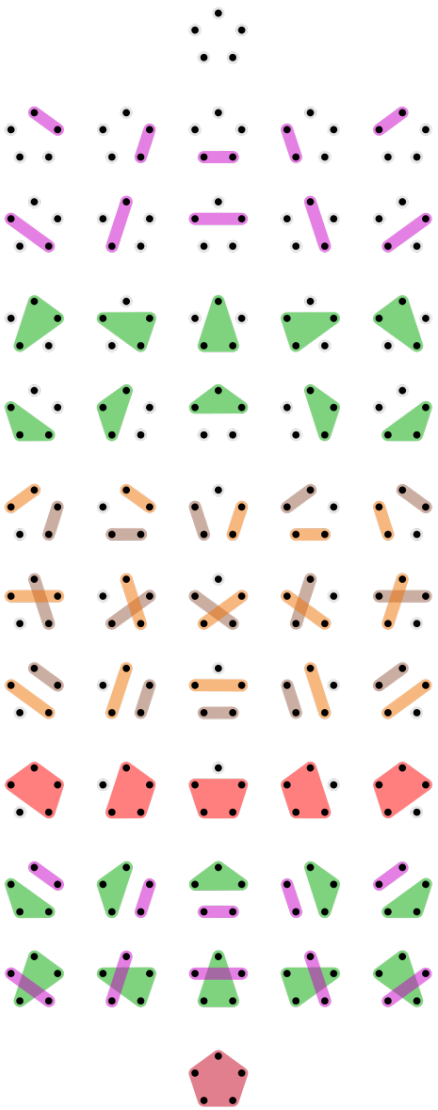
$\mathcal{P}(S)$ 上的包含 $\subseteq$关系

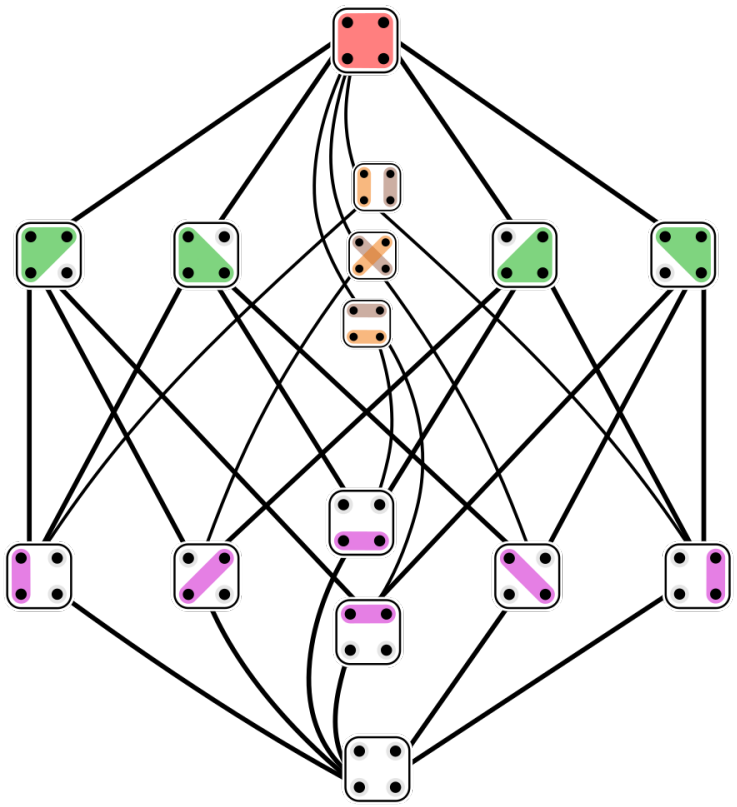


自反 + 反对称 + 传递

B_n: Boolean algebra of degreen n

Partitions of a Set; Ordered by **Refinement (细化/精化)**





$$\alpha \leq \beta \iff \forall A \in \alpha\ \exists B \in \beta\ A \subseteq B$$

Definition (严格偏序关系 (Strict Partial Order))

令 $\prec \subseteq X \times X$ 是 X 上的二元关系。

如果 $\prec$ 满足以下条件, 则称 $\prec$ 是 X 上的严格偏序关系:

(1) $\prec$ 是反自反 (irreflexive)的。

$$\forall x \in X \ \neg(x \prec x).$$

(2) $\prec$ 是**非对称 (asymmetric)**的。

$$\forall x, y \in X \ x \prec y \to \neg(y \prec x).$$

(3) $\prec$ 是传递 (transitive)的。

$$\forall x, y, z \in X \ x \prec y \wedge y \prec z \to x \prec z.$$

$$(\mathcal{P}(X), \subset)$$

$$(1) + (3) \implies (2) \qquad \text{反对称?}$$

Theorem

设 $\prec \subseteq X \times X$ 是 X 上的严格偏序关系。

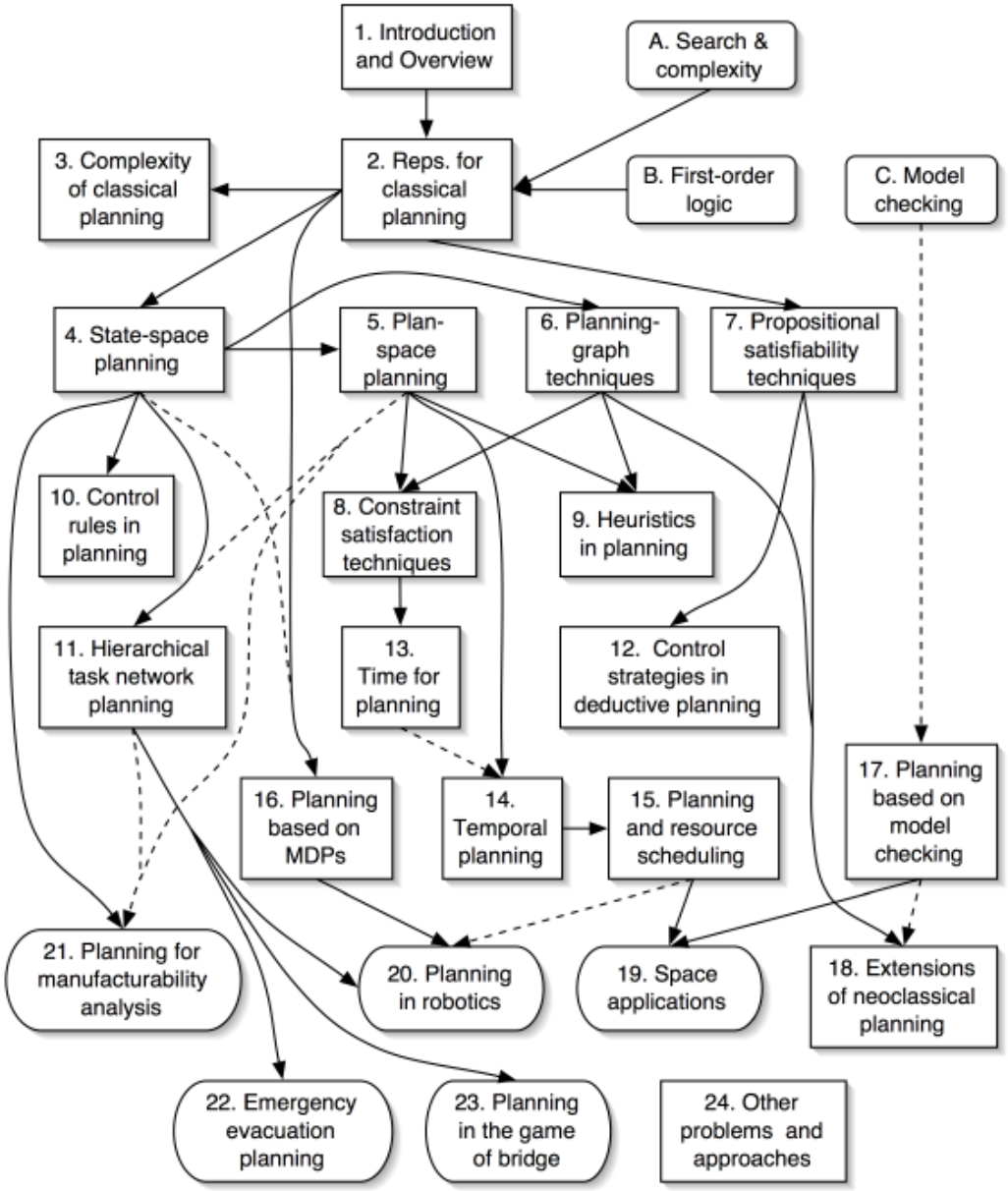
对于任意 $x, y, z \in X$:

(1) $x \prec y$, $x = y$, $y \prec x$ 三者中至多有一个成立;

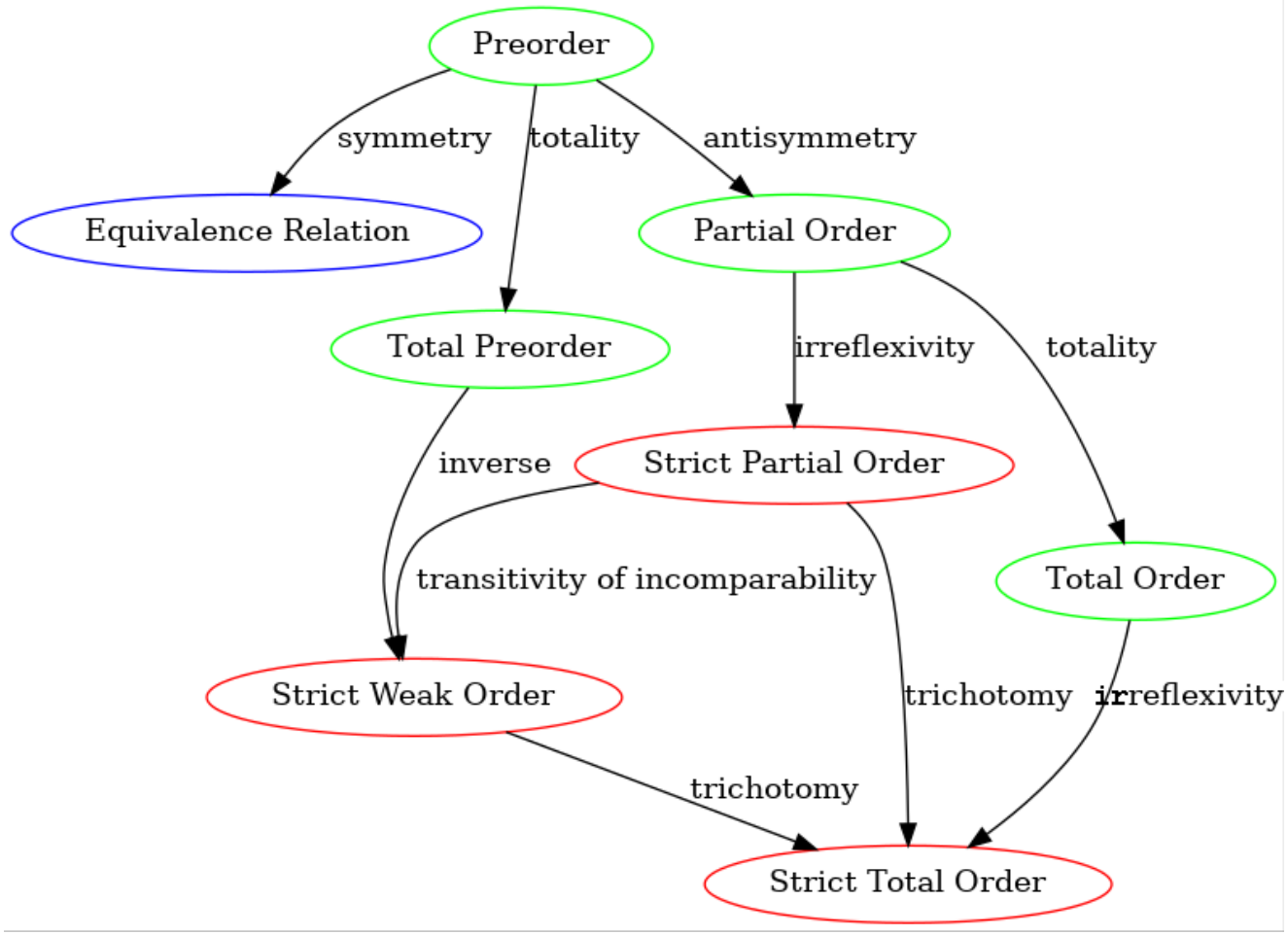
(2) $(x \preceq y \wedge y \preceq x) \to x = y$。

$$(x \preceq y \triangleq x \prec y \vee x = y)$$

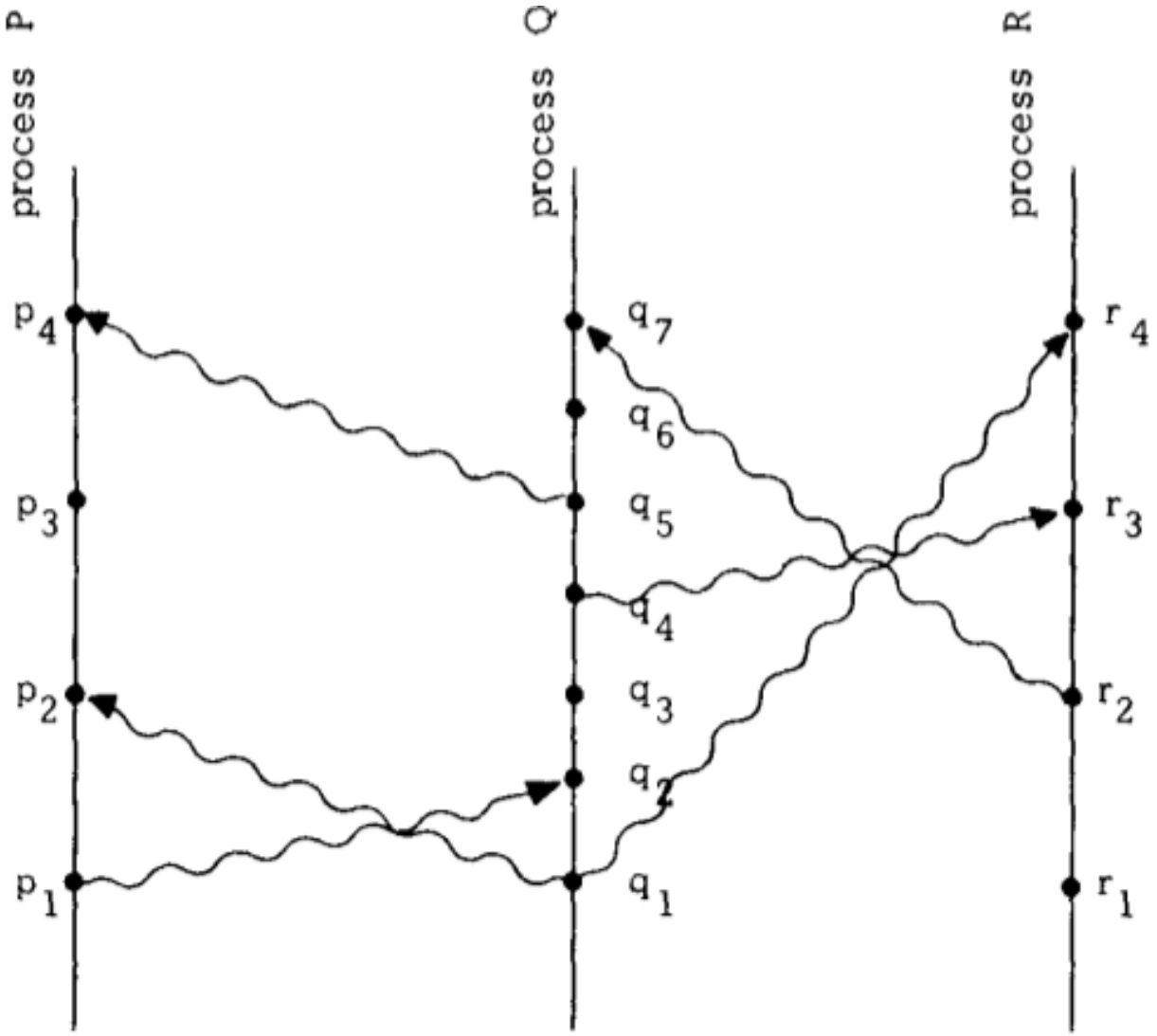
$$\begin{aligned}
& x \preceq y \wedge y \preceq x \\
\iff & (x \prec y \vee x = y) \wedge (y \prec x \vee x = y) \\
\iff & x = y \vee (x \prec y \wedge y \prec x) \\
\implies & x = y \vee \text{False} \\
\implies & x = y
\end{aligned}$$


1. Introduction and Overview
A. Search & complexity
3. Complexity of classical planning
2. Reps. for classical planning
B. First-order logic
C. Model checking
4. State-space planning
5. Plan-space planning
6. Planning-graph techniques
7. Propositional satisfiability techniques
10. Control rules in planning
8. Constraint satisfaction techniques
9. Heuristics in planning
11. Hierarchical task network planning
13. Time for planning
12. Control strategies in deductive planning
17. Planning based on model checking
16. Planning based on MDPs
14. Temporal planning
15. Planning and resource scheduling
21. Planning for manufacturability analysis
20. Planning in robotics
19. Space applications
18. Extensions of neoclassical planning
22. Emergency evacuation planning
23. Planning in the game of bridge
24. Other problems and approaches

反自反 + 传递


Preorder
symmetry
totality
antisymmetry
Equivalence Relation
Partial Order
Total Preorder
irreflexivity
totality
inverse
Strict Partial Order
transitivity of incomparability
Total Order
Strict Weak Order
trichotomy
irreflexivity
trichotomy
Strict Total Order

反自反 + 传递


process P
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process R
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p3
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q7
q6
q5
q4
q3
q2
q1
r4
r3
r2
r1

Definition (事件间的 “先于” 关系 (“Happened Before” Relation on Events))

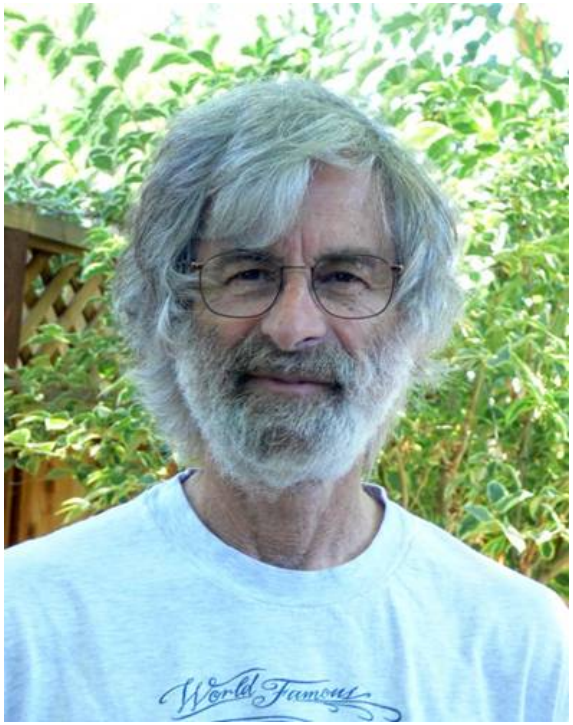
令 E 是表示事件的集合。

$\rightarrow \subseteq E \times E$ 是 E 上的“先于”关系, 如果它是满足以下条件的最小关系:

- 对任何事件 $a \in E$, $\neg(a \rightarrow a)$;
- 如果 a, b 事件属于同一个进程且 a 在 b 之前发生, 则 $a \rightarrow b$;
- 如果 a, b 分别表示同一个消息的发送事件与接收事件, 则 $a \rightarrow b$;
- 如果 $a \rightarrow b$ 且 $b \rightarrow c$, 则 $a \rightarrow c$。

Theorem

$\rightarrow \subseteq E \times E$ 是 E 上的严格偏序。



Leslie Lamport (1941 ∼)

For "fundamental contributions to the theory and practice of **distributed and concurrent systems**, *notably the invention of concepts such as causality and logical clocks, safety and liveness, replicated state machines, and sequential consistency"*

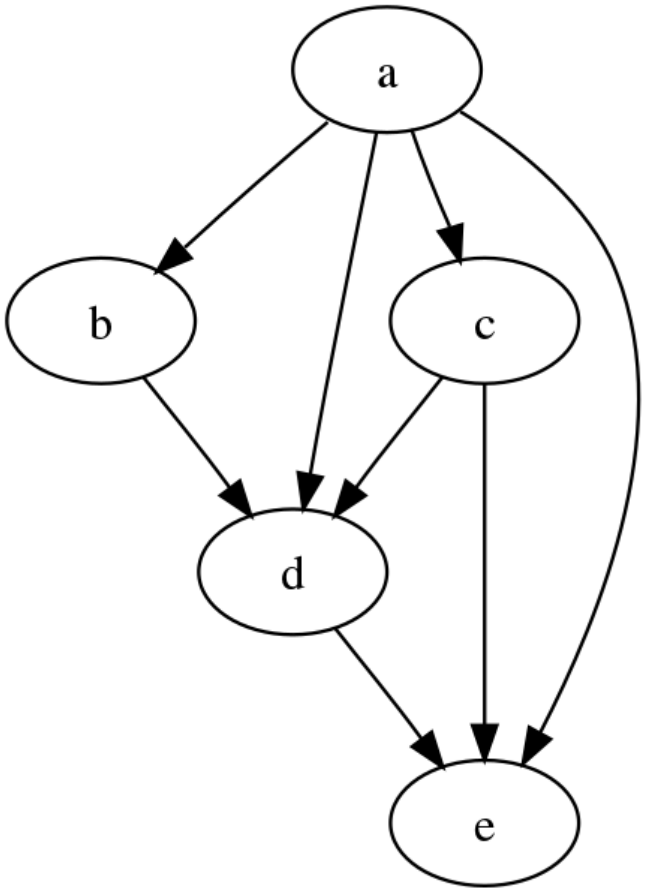
— *Turing Award (2013)*

Time, Clocks, and the Ordering of Events in a Distributed System

Leslie Lamport
Massachusetts Computer Associates, Inc.

The concept of one event happening before another in a distributed system is examined, and is shown to define a partial ordering of the events. A distributed algorithm is given for synchronizing a system of logical clocks which can be used to totally order the events.

引用量 > 14600



有向无环图 (DAG; Directed Acyclic Graph)

Definition (全序关系 (Total Order))

如果 $\preceq \subseteq X \times X$ 满足以下性质, 则称 $\preceq$ 是 X 上的全序关系:

(1) 自反 (Reflexivity):

$$\forall x \in X \; x \preceq x$$

(2) 反对称性 (Antisymmetric):

$$\forall x, y \in X \; (x \preceq y \wedge y \preceq x) \to x = y$$

(3) 传递性 (Transitive):

$$\forall x, y, z \in X \; x \preceq y \wedge y \preceq z \to x \preceq z$$

(4) 连接性 (Connex; Totality):

$$\forall x, y \in X \; x \preceq y \vee y \preceq x.$$

全序关系 = 偏序关系 + 连接性 $\qquad (4) \implies (1)$

全序关系 = 偏序关系 + 连接性

$$(\mathbb{R}, \leq) \qquad (\mathbb{R}, \geq)$$

Definition (严格全序关系 (Strict Total Order))

如果 $\prec \subseteq X \times X$ 满足以下条件, 则称 $\prec$ 是 X 上的严格全序关系:

(1) 反自反 (Irreflexivity):

$$\forall x \in X \ \neg(x \prec x)$$

(2) 传递性 (Transitive):

$$\forall x, y, z \in X \ x \prec y \wedge y \prec z \rightarrow x \prec z$$

(3) 半连接性 (Semi-Connex):

$$\forall x, y \in X \ \neg(x = y) \rightarrow ((x \prec y) \vee (y \prec x))$$

$$(\mathbb{R}, <) \qquad (\mathbb{R}, >)$$

严格全序关系 = 严格偏序关系 + 半连接性

Theorem

设 $\prec \subseteq X \times X$ 是 X 上的严格全序关系。

$\prec$ 满足三歧性 *(Trichotomous)*:

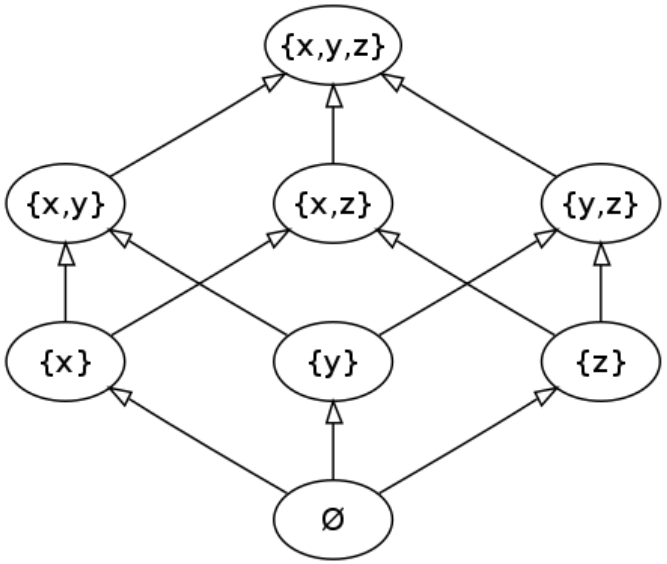
$$\forall x, y \in X \ (\textit{exactly one of}\ x \prec y, x = y,\ \textit{or}\ y \prec x\ \textit{holds})$$

Definition

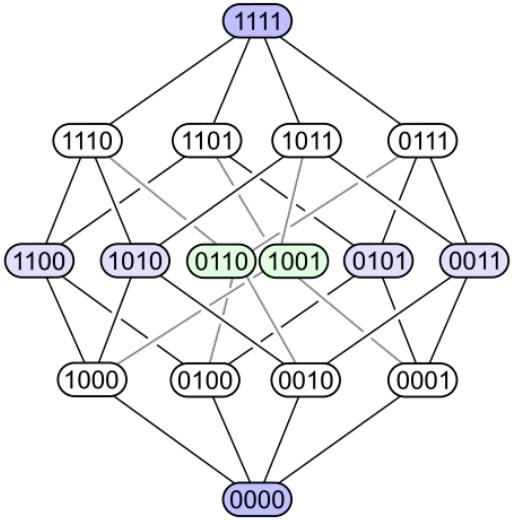
设 $(X, \preceq)$ 是偏序集。对任意 $a, b \in X$, 如果

$$a \prec b \wedge (\forall c \in X\ (c \neq a \wedge c \neq b) \rightarrow \neg(a \preceq c \preceq b))$$

则称 a 被 b 覆盖 (covered)。



哈斯图 (Hasse diagram)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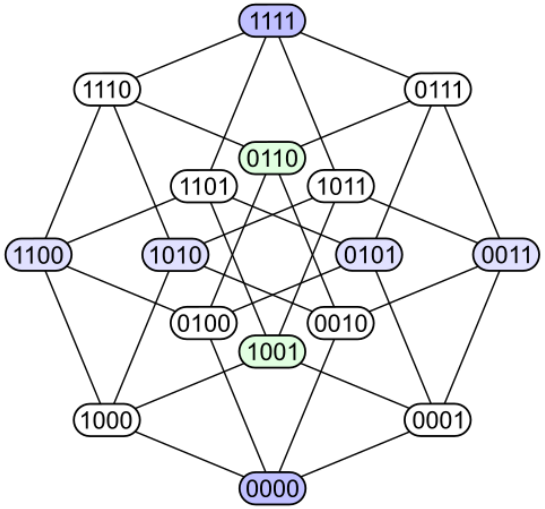
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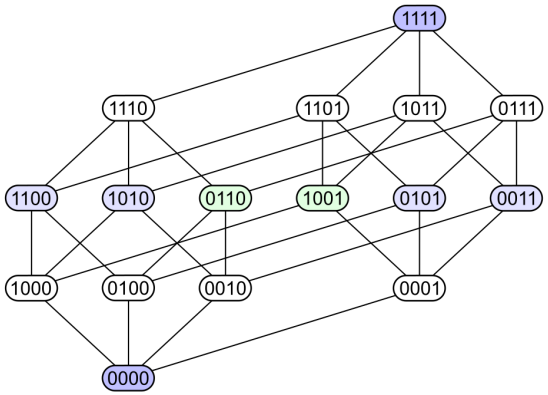
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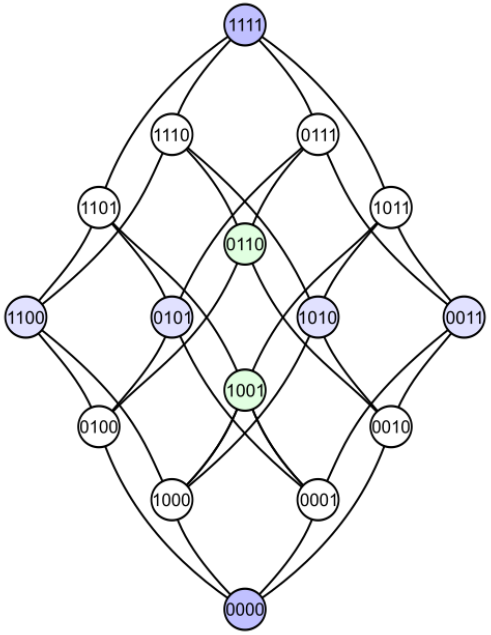
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Definition (极大元与极小元 (Maximal/Minimal))

令 $\preceq \subseteq X \times X$ 是 X 上的偏序。设 $a \in X$。

如果

$$\forall x \in X \ \neg(a \prec x),$$

则称 a 是 X 的极大元。

如果

$$\forall x \in X \ \neg(x \prec a),$$

则称 a 是 X 的极小元。

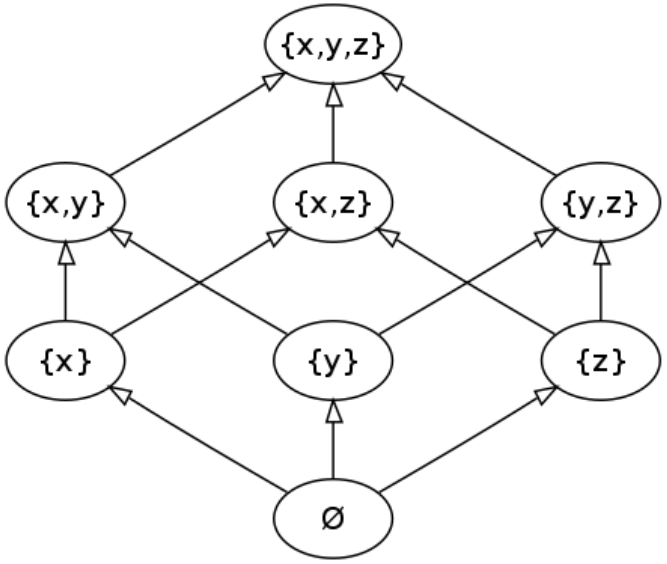
Q：极大/极小元是否一定存在？如果存在，是否唯一？

$$(\mathbb{R}, \leq)$$

无极大元、无极小元

$$(\mathbb{N}, \leq)$$

无极大元、有唯一极小元 0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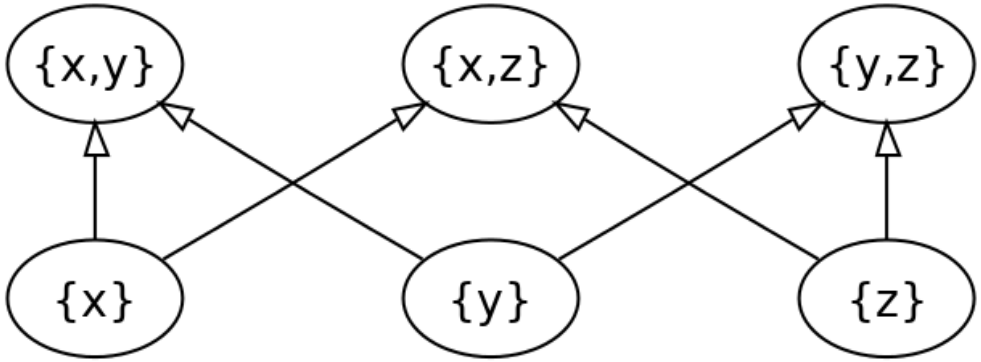
{x,y}
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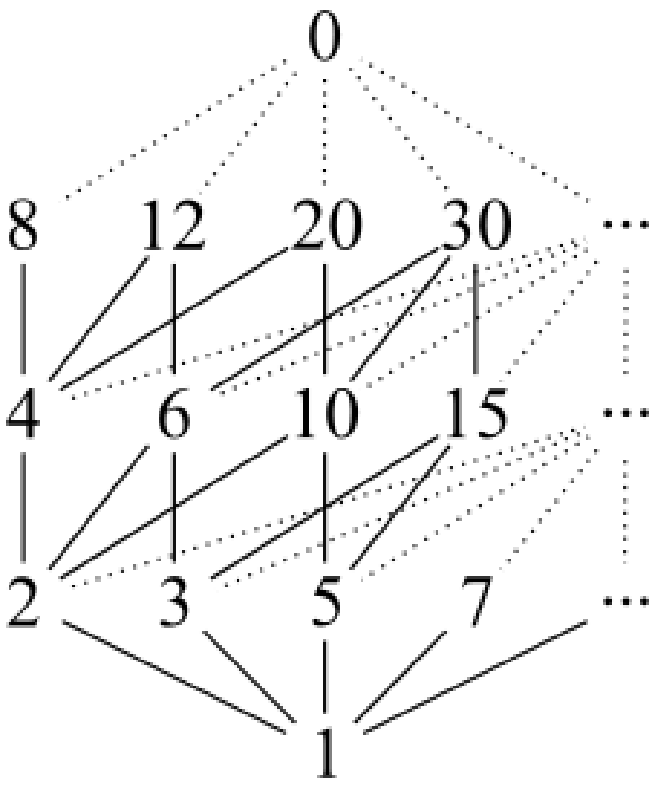
{y,z}
{x}
{y}
{z}

$$(\{2,3,4,\dots\} = \mathbb{N} \setminus \{0,1\}, |)$$



无极大元、有无穷多个极小元 (所有的素数)

Definition (最大元与最小元 (Maximum/Minimum; Greatest/Least; Largest/Smallest))

令 $\preceq \subseteq X \times X$ 是 X 上的偏序。设 $a \in X$。

如果

$$\forall x \in X \; x \preceq a,$$

则称 a 是 X 的最大元。

如果

$$\forall x \in X \; a \preceq x,$$

则称 a 是 X 的最小元。

Q：最大/最小元是否一定存在？如果存在，是否唯一？

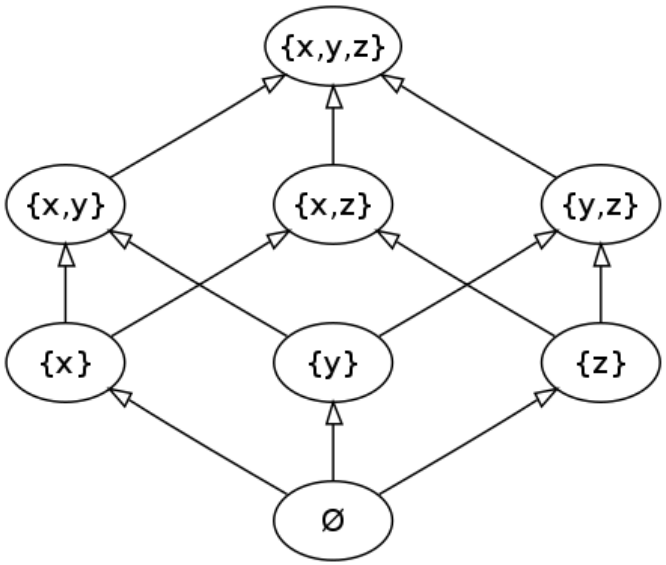
Theorem

偏序集 $(X, \preceq)$ 如果有最大元或最小元，则它们是唯一的。

假设存在两个最大元 x 与 y。

$$x \preceq y \wedge y \preceq x \implies x = y$$

(反对称性)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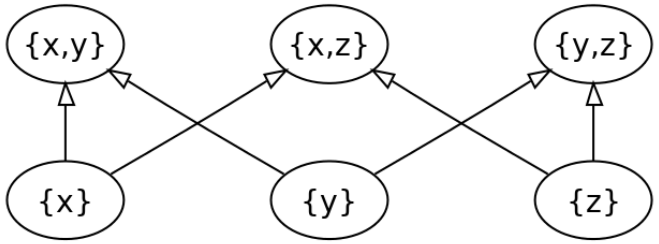
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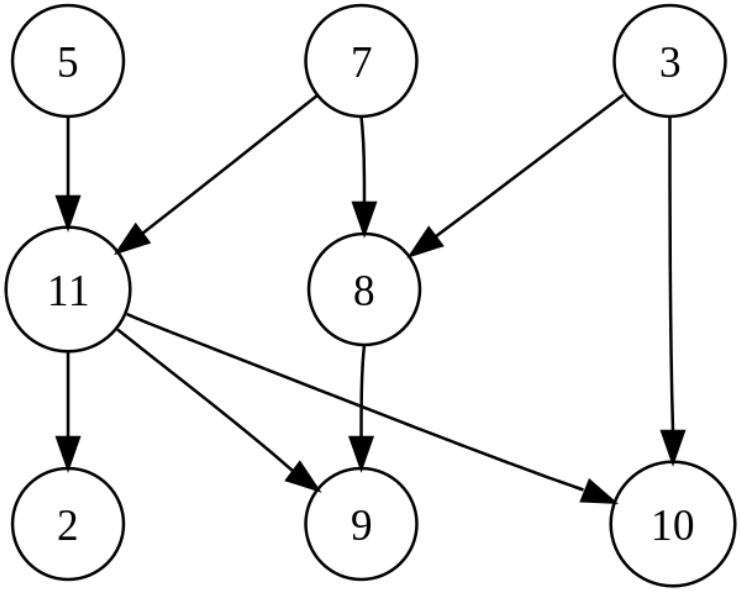
{z}

Definition (线性拓展 (Linear Extension))

设 $(X, \preceq)$ 是偏序集, $(X, \preceq')$ 是全序集。如果

$$\forall x, y \in X \; x \preceq y \to x \preceq' y,$$

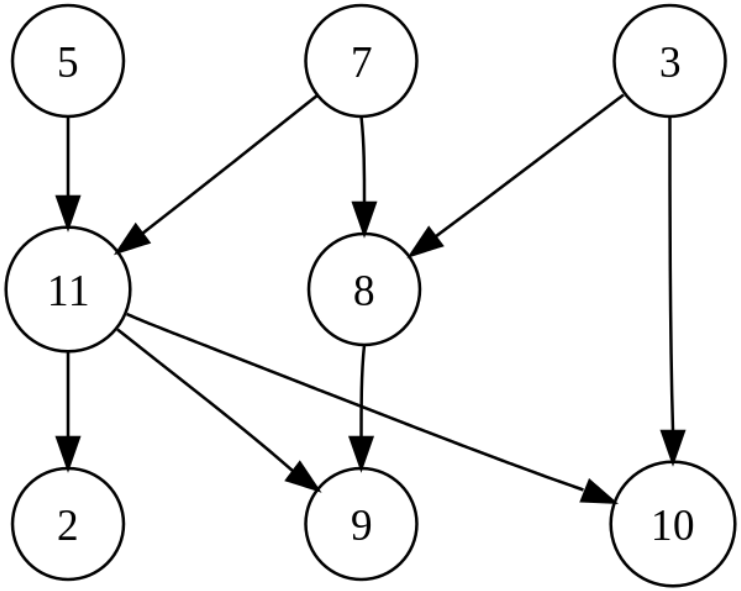
则称 $\preceq'$ 是 $\preceq$ 的线性拓展。



$5, 7, 3, 11, 8, 2, 9, 10$ $\qquad$ $3, 5, 7, 8, 11, 2, 9, 10$

Theorem

设 $(X, \preceq)$ 是偏序集且 X 是有限集，则 $\preceq$ 的线性拓展必定存在。



Topological sorting: iteratively remove minimal elements

Theorem

设 $(X, \preceq)$ 是偏序集且 X 是有限集，则极小元 (极大元) 一定存在。

Definition (上界 (Upper Bound) 与下界 (Lower Bound))

设 $(X, \preceq)$ 是偏序集。对于 $Y \subseteq X$, 如果

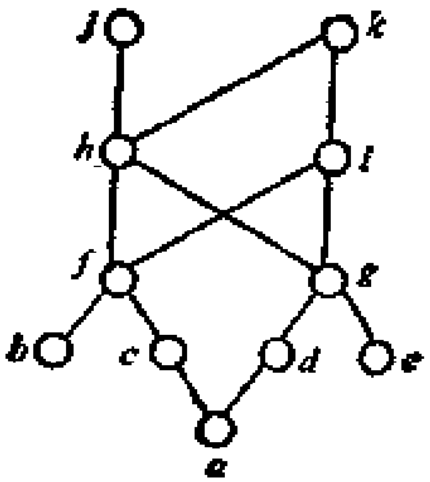
$$\exists x \in X \ \forall y \in Y \ y \preceq x,$$

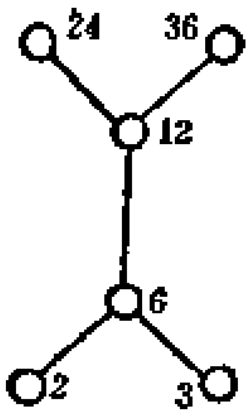
则称 x 为 Y 的上界。

类似地, 如果

$$\exists x \in X \ \forall y \in Y \ x \preceq y,$$

则称 x 为 Y 的下界。

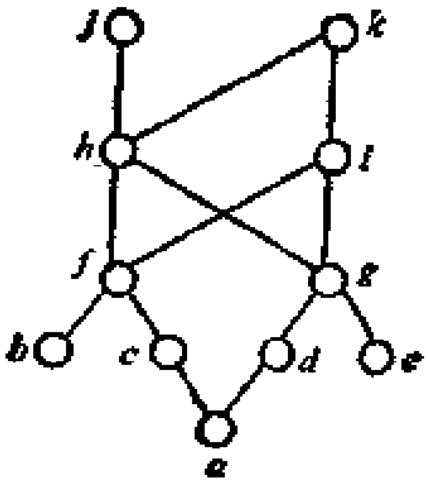


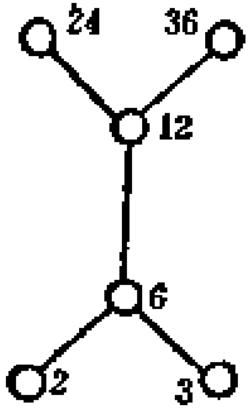


Definition (最小上界 (Least Upper Bound; LUB))

设 $(X, \preceq)$ 是偏序集, $Y \subseteq X$ 是 X 的子集, $x \in X$ 是 Y 的上界。

若对于 Y 的所有上界 x', 均有 $x \preceq x'$, 则称 x 是 Y 的最小上界/上确界。





Definition (最大下界 (Greatest Lower Bound; GLB))

设 $(X, \preceq)$ 是偏序集, $Y \subseteq X$ 是 X 的子集, $x \in X$ 是 Y 的下界。

若对于 Y 的所有下界 x', 均有 $x' \preceq x$, 则称 x 是 Y 的最大下界/下确界。

如下 $(A, \leq)$ 有上确界吗?

$$A = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 < 2\}$$

在 $(\mathbb{R}, \leq)$ 中, A 有上确界: $\sup(A) = \sqrt{2}$

在 $(\mathbb{Q}, \leq)$ 中, A 无上确界

Theorem

集合 A 在 $(\mathbb{Q}, \leq)$ 中无上确界。

$$A = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 < 2\}$$

Proof.

$$B = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 > 2\}$$

只需证明: A 无最大元、B 无最小元

$$\forall p \in A\ \exists q \in A\ p < q \qquad \forall p \in B\ \exists q \in B\ p > q$$

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} \qquad q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}$$

$$p \in A \implies q \in A \wedge p < q \qquad p \in B \implies q \in B \wedge p > q$$

$\square$

Definition (实数 ($\mathbb{R}$))

如果集合 $A \subseteq \mathbb{Q}$ 满足:

(1) $A \neq \emptyset$ 且 $A \neq \mathbb{Q}$;

(2) A 是向下封闭的:

$$\forall p \in A, q \in \mathbb{Q}\ (q < p \to q \in A)$$

(3) A 没有最大元:

$$\forall p \in A\ \exists q \in A\ p < q$$

则称 A 是**戴德金分割** (Dedekind cut), A 所对应的实数为 $\sup(A)$。

Definition ($\leq_{\mathbb{R}}$)

$x_1 \leq_{\mathbb{R}} x_2 \iff x_1 \subseteq x_2$

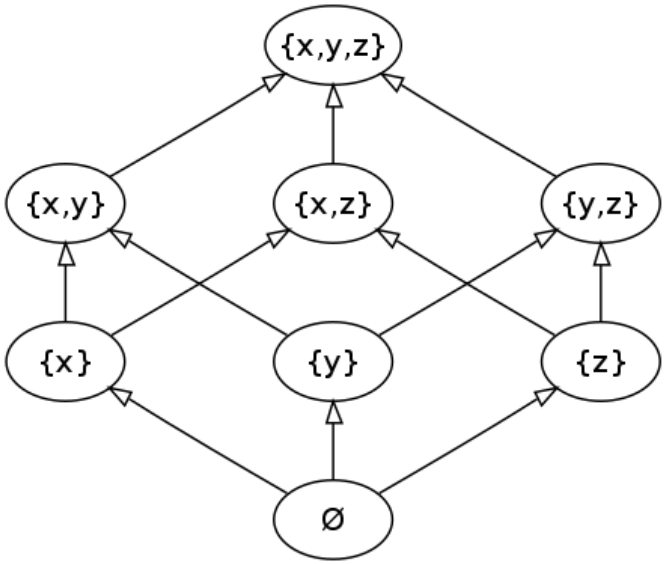
Definition ($+_{\mathbb{R}}$)

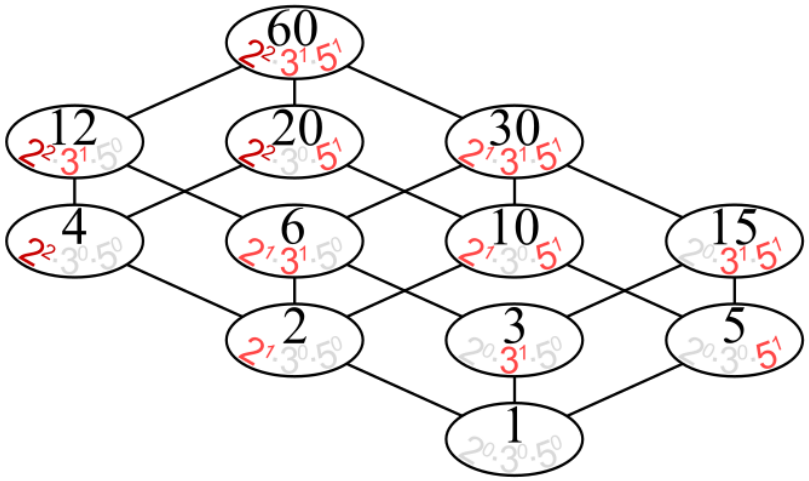
$x +_{\mathbb{R}} y = \{p +_{\mathbb{Q}} q \mid p \in x, q \in y\}$

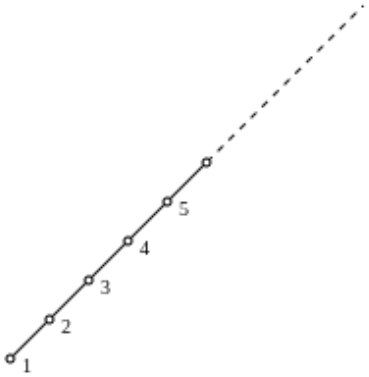
Definition (格 (Lattice))

设 $(X, \preceq)$ 是偏序集。

如果任意两个元素都有最小上界与最大下界, 则称 $(X, \preceq)$ 为格。







"LET'S STOP HERE, SO WE HAVE
ONLY GOOD MEMORIES
OF WHAT WE LIVE."
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Definition (良序 (Well-Ordering))

设 $(X, \preceq)$ 是偏序集。

如果 X 的任意非空子集都有最小元, 则称 $(X, \preceq)$ 为良序集。

$$(\mathbb{N}, \leq) \qquad (\mathbb{Z}, \leq) \qquad (\mathbb{R}, \leq)$$

数学归纳法　　良序集

Theorem

良序集 $(X, \preceq)$ 一定是全序集。

对任意两个元素 $x, y \in X$, 考虑 $\{x, y\}$ 非空子集。

$$x \preceq y \vee y \preceq x$$

Theorem

每个有限的全序集 $(X, \preceq)$ 一定是良序集。

$$X = \{x_1, x_2, \ldots, x_n\}$$

反设 $(X, \preceq)$ 不是良序集。

存在非空子集 $\emptyset \neq Y \subseteq X$, Y 中没有最小元素。

则 Y 中存在不可比元素　(Y 是有限的!)

Definition

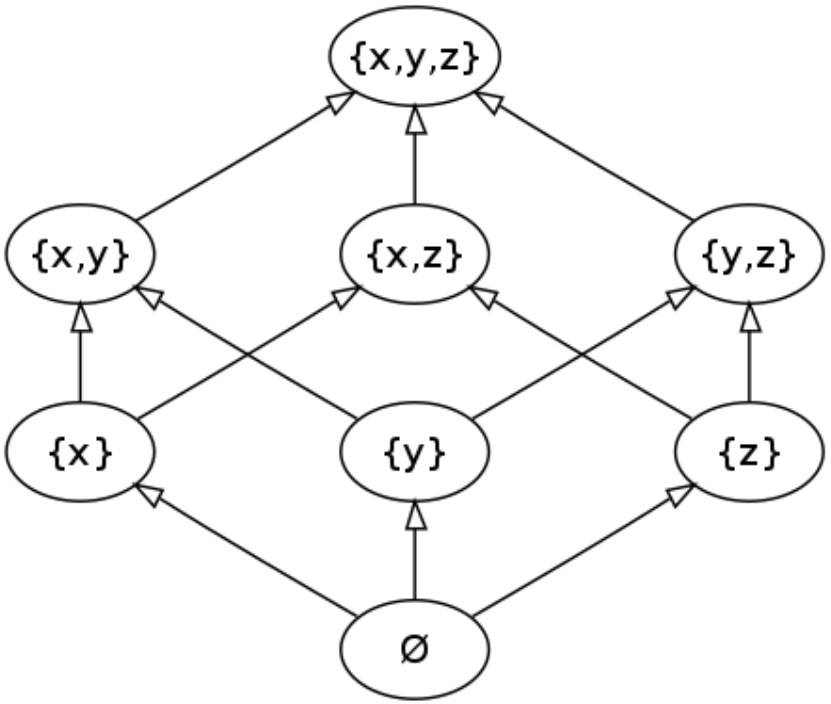
设 $(X, \preceq)$ 是偏序集。对 $a, b \in X$,

可比的 (Comparable):

$$a \preceq b \vee b \preceq a$$

不可比的 (Incomparable):

$$\neg(a \preceq b \vee b \preceq a)$$

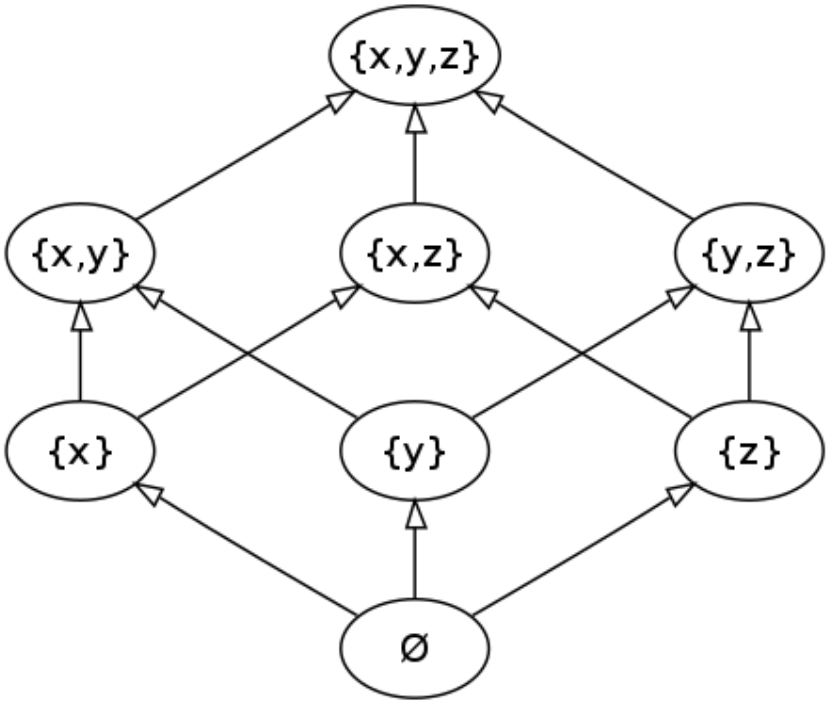


Definition (链与反链 (Chain; Antichain))

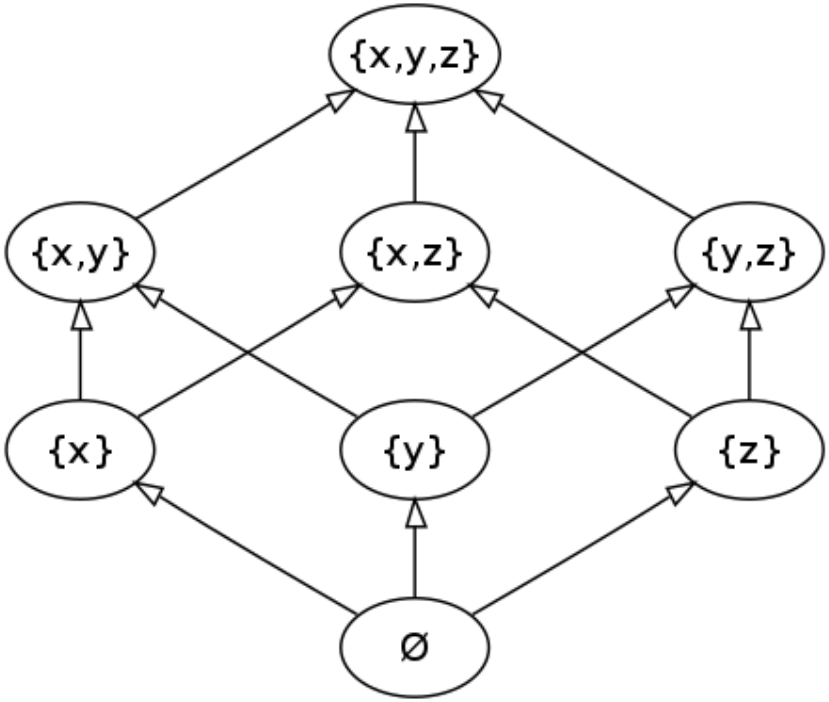
设 $(X, \preceq)$ 是偏序集。

- 设 $S \subseteq X$ 且 S 中元素两两可比, 则称 S 是链。
- 设 $S \subseteq X$ 且 S 中元素两两不可比, 则称 S 是反链。

规定: 单元素集合既是链, 也是反链



$$\{\{\emptyset, \{x\}, \{x, y\}, \{x, y, z\}\}, \{\{y\}, \{y, z\}\}, \{\{z\}, \{x, z\}\}\}$$

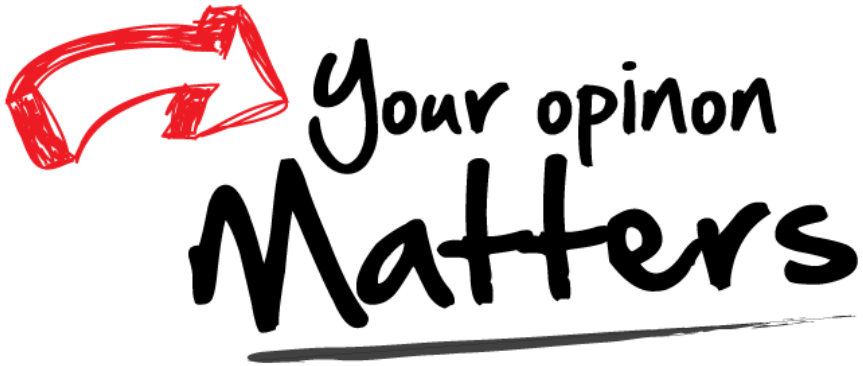


$$\{\{x\}, \{y\}, \{z\}\}$$

Theorem (Dilworth's Theorem)

最大反链的大小 = 最小链分解中链的条数

Thank You!



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